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Research Article

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Tipping as a Pricing Signal

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Abstract: This paper proposes a menu-cost theory of tipping. When firms face costs of changing posted prices, voluntary tips provide a noisy but informative signal of whether current prices are "too low" or "too high." The firm adjusts its price only when tip deviations exceed an information-dependent threshold, generating a time-varying inaction band. The framework explains why tipping is common in volatile, service-intensive industries but rare in standardized chains with rich alternative telemetry. Unlike existing models based on norms or reciprocity, this approach requires only that average tips increase weakly with perceived surplus for the consumers.

Keywords: tipping; menu costs; state-dependent pricing; information

JEL Classification: D21; D83; L11

1 Introduction

Tipping behavior is often attributed to social norms or altruism on the part of consumers. This paper instead studies tipping from the suppliers' side and asks how voluntary tips can be useful when posted prices are costly to change. The paper shows that tipping can operate as an informational device in a standard menu-cost pricing problem: tips provide a noisy signal of mispricing, and the optimal pricing policy features a state-dependent band of inaction in which small shocks are absorbed by tips while sufficiently large deviations trigger discrete posted-price changes. Unlike existing models based on norms or reciprocity, the mechanism requires only that average tips increase weakly with perceived surplus.

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The model helps rationalize a fact emphasized in the literature: tipping is prevalent in full-service, high-variation environments and uncommon in standardized chains with centralized pricing. The model can also help us understand a more recent phenomenon: digital checkout systems increasingly prompt customers to tip before any service is provided. Rather than treating this as a mere shift in norms, such early tipping can be viewed as evidence that firms value tips as immediate feedback about pricing or perceived surplus – a central idea in the model below.

Before developing the formal model, it is useful to state the central idea in plain language. Consider a restaurant that prints its menu prices and cannot change them costlessly. Between menu reprints, the restaurant observes tips: if customers consistently tip well above the norm, this suggests that the current price is a bargain relative to the value customers perceive; the price may be "too low." Conversely, persistently low tips suggest overpricing. The restaurant uses this tip signal to decide whether the informational benefit of reprinting the menu (better-aligned prices) justifies the cost of doing so. The formal model makes this logic precise and derives its implications.

1.1 Relation to the Tipping Literature

Most economic analyses of tipping emphasize social norms, reciprocity, or fairness motives and study their implications for service quality, compensation, or welfare (e.g., Azar (2007, 2020); Lynn (2015)). A large empirical literature documents that tip amounts respond to customers' perceptions of value and satisfaction (Whaley et al. 2019), that the economic value of tipping to restaurants depends on customer price sensitivity (Lynn 2017), and that tips at the establishment level predict future sales (Lynn and Ukhov 2013). These findings are consistent with tips carrying information about pricing adequacy. On the supply side, Schwartz (1997) models tipping as a form of price discrimination, and Lynn and Wang (2013) show that tipping policies affect customers' perceptions of expensiveness and fairness, implying that customers treat the posted price and the expected tip as joint components of total cost, a pattern consistent with the channel through which tips convey surplus information in the model below.

This paper's contribution is orthogonal to these approaches: I treat tips primarily as an informational signal that helps firms infer mispricing when posted prices are costly to change. On the demand side I assume only that the average tip rate weakly increases with perceived surplus, and I allow substantial noise from defaults or context (e.g., Haggag and Paci 2014); I do not rely on a particular norm-formation process, strong reciprocity preferences, or other behavioral assumptions. The empirical finding that tips increase with perceived value

(Whaley et al. 2019), together with the observation that tipping's economic value to restaurants depends on customer price sensitivity (Lynn 2017), is consistent with this reduced-form assumption; the model does not require that the relationship be strong or precise, only that it exists on average. On the supply side, combining signal extraction with menu costs yields a time-varying, information-dependent band of inaction and cross-sectional predictions about when firms will use or ignore tipping (e.g., substitution toward internal telemetry in large chains). The finding that tips predict restaurant sales (Lynn and Ukhov 2013) is consistent with the model's premise that tips contain forward-looking information relevant to the firm's pricing decisions. In this way, the paper complements norm-based and incentive-based accounts by identifying a distinct pricing-information role for tipping. Relative to Schwartz (1997), who treats tips as enabling price discrimination across heterogeneous consumers, the present paper focuses on the intertemporal signal-extraction problem: tips help the firm learn about changes in the common efficient price over time, not about cross-sectional differences across consumers.

2 Model

This section develops the formal model in four steps. First, I set up the pricing environment with menu costs and a tip-based signal of mispricing (Section 2.1 presents comparative statics). Then Section 2.2 shows that the informational mechanism does not require an exogenous tipping norm. Section 2.3 extends the framework to a competitive setting, and Section 2.4 embeds the static rule in a dynamic environment where beliefs evolve over time. Section 3 collects the testable implications.

I begin with a standard state-dependent pricing in the style of Caplin and Spulber (1987). Time is discrete. A firm sells a good or provides a service at posted price p_t.[1] The latent efficient price p_t^* evolves as

$$p_t^* = p_{t-1}^* + \varepsilon_t, \quad \varepsilon_t \sim N\left(0, \sigma_\varepsilon^2\right). \tag{1}$$

Changing p_t entails a menu cost $\kappa > 0$. Demand is $Q(p)$ with standard properties, and instantaneous profit is $\pi(p) = (p - c)Q(p)$ with $\pi''(p^*) < 0$. Consumers may leave a voluntary tip $t_t \geq 0$, summarized by a percentage $\tau_t \equiv t_t/p_t$. I assume a reduced-form measurement equation

$$\tau_t - \bar{\tau} = \alpha\left(p_t^* - p_t\right) + \eta_t, \quad \eta_t \sim N\left(0, \sigma_\eta^2\right),\ \alpha > 0, \tag{2}$$

1 In the context of a restaurant, it is often a combination of goods and services. The signal extraction problem described in the model may apply to the firm as a whole, or only to specific goods or services sold by the firm.

i.e., tips rise with perceived surplus $p_t^* - p_t$ up to noise η_t. The parameter $\bar{\tau}$ is the average level of tip (which can be determined by social norm), and serves here as a convenient reduced-form reference level. The linear–Gaussian structure in (2) is adopted for transparency and closed-form updating; the menu-cost pricing rule below depends only on the posterior mean mispricing, and therefore extends to more general signal structures. While $\bar{\tau}$ is taken as given in the baseline specification, Section 2.2 provides a robustness extension in which one can dispense with an exogenous tipping norm: with customer search frictions, the mean tipping rate is pinned down endogenously while the menu-cost pricing logic remains unchanged. Notice that the assumptions I impose on consumers are relatively weak: I only require the tip rate to increase (possibly weakly) with perceived surplus, with some noise added in. The parameter α can be small and the signal can be noisy, and the firm can still learn from observing tips.

The model rests on three key assumptions. First, there is an "efficient" or target price p_t^* that the firm would like to charge, but this target shifts over time in ways the firm cannot perfectly observe (e.g., due to changing demand conditions, input costs, or competitor actions). Second, changing the posted price is costly; the firm must pay a menu cost κ each time it updates its price. Third, tips are informative: when customers perceive that they are getting a good deal (price is below the target), they tend to tip more, and when they perceive the price as too high, they tend to tip less. This last assumption is supported by the empirical evidence in Whaley et al. (2019) and Lynn (2017).

Information structure and timing. At the start of period t the firm has a posted price p_t in place and a prior over the mispricing $s_t \equiv p_t^* - p_t$. The efficient price p_t^* is realized but not observed by the firm. Consumers experience (and tip according to) their perceived surplus; the firm observes the realized tip rate τ_t (and sales) before choosing whether to incur the menu cost κ and change its posted price. Upon observing τ_t the firm updates beliefs about s_t via Bayes' rule and then chooses an adjustment Δ_t (interpretable as an immediate change or as the update applied for the next pricing decision). Formally, the one-period problem below conditions on the realized signal when evaluating the adjustment decision.

Given a Gaussian prior $s_t \sim N(\mu_{t|t-1}, v_{t|t-1})$ and the signal $\tau_t - \bar{\tau} = \alpha s_t + \eta_t$, $\eta_t \sim N\left(0, \sigma_\eta^2\right)$, the posterior mean is

$$m_t \equiv \mathbb{E}[s_t \mid \tau_t] = (1 - \alpha\lambda_t)\, \mu_{t|t-1} + \lambda_t(\tau_t - \bar{\tau}), \qquad \lambda_t \equiv \frac{\alpha\, v_{t|t-1}}{\alpha^2 v_{t|t-1} + \sigma_\eta^2}, \tag{3}$$

and the posterior variance updates to $v_t = (1 - \alpha\lambda_t)v_{t|t-1}$.

A weak or noisy tip signal is not as useful as a strong one. If α is small or σ_η^2 is high, the firm puts less weight on any single tip realization because each observed tip is a less reliable guide to mispricing. Learning still occurs, but it occurs more slowly. In practical terms, the firm waits for larger or more persistent tip deviations before concluding that the posted price is meaningfully out of line and worth changing. The mechanism survives as long as tips contain some information on average. Only if tips are completely unrelated to mispricing would they become useless for pricing decisions.

Unless otherwise noted, I adopt the normalization $\mu_{t|t-1} = 0$ after any price adjustment (so the prior is re-centered at zero); during inaction, $\mu_{t|t-1}$ generally need not be zero, in which case the inaction region in tip space is centered at $\tau_t = \bar{\tau} - \frac{1-\alpha\lambda_t}{\lambda_t}\mu_{t|t-1}$ rather than at $\bar{\tau}$.

Definition 1 (Stationary pricing equilibrium). In this partial-equilibrium environment, a stationary pricing equilibrium consists of (i) a time-invariant pricing (adjustment) policy mapping the firm's posterior beliefs into an adjustment decision and size, and (ii) a belief-updating rule consistent with Bayes' rule under the tip signal. Given the exogenous process for p_t^*, the policy and updating rule induce a stationary distribution for mispricing s_t (and hence for tips).

Loss from mispricing. Quadratically approximating around p^* gives expected static loss from mispricing $\ell(s) = \frac{1}{2}\Lambda s^2$ where $\Lambda \equiv -\pi''(p^*) > 0$. If the firm *does not* adjust, its expected loss is $\mathbb{E}[\ell(s_t) \mid \tau_t] = \frac{1}{2}\Lambda\left(m_t^2 + v_t\right)$. If the firm *does* adjust, it sets $p_t \leftarrow p_t + m_t$ (Bayes-optimal myopic move), leaving residual variance v_t and expected loss $\frac{1}{2}\Lambda v_t$. The expected *gain* from adjusting is therefore

$$G_t \equiv \frac{1}{2}\Lambda m_t^2. \tag{4}$$

Proposition (Inaction band and tip-triggered pricing). *Let $s_t \equiv p_t^* - p_t$ denote mispricing. Suppose the firm faces a per-change menu cost $\kappa > 0$ and a one-period quadratic loss from mispricing $\ell(s) = \frac{1}{2}\Lambda s^2$ with $\Lambda > 0$. At the beginning of period t the firm observes a signal τ_t (e.g., tips) and forms a posterior for s_t with mean $m_t \equiv \mathbb{E}[s_t \mid \tau_t]$ and variance $v_t \equiv \mathrm{Var}(s_t \mid \tau_t)$. If the firm changes its posted price by $\Delta_t \in \mathbb{R}$ it pays κ and the period-t mispricing becomes $s_t - \Delta_t$; otherwise $\Delta_t = 0$. Then the optimal one-period policy is:*

(i) *Conditional on adjusting, the unique optimal size is $\Delta_t^* = m_t$.*
(ii) *The firm adjusts if and only if $\frac{1}{2}\Lambda m_t^2 > \kappa$.*

In particular, there is a band of inaction for the posterior mean mispricing $m_t \in [-s^*, +s^*]$.

If, in addition, the mapping $\tau_t \mapsto m_t$ is strictly monotone, the rule can be equivalently stated as a two-sided threshold in the signal space: $|m(\tau_t)| > s^*$. Under the linear–Gaussian measurement equation $\tau_t - \bar{\tau} = \alpha s_t + \eta_t$, with prior

$s_t \sim N(\mu_{t|t-1}, v_{t|t-1})$ and $\eta_t \sim N\left(0, \sigma_\eta^2\right)$ independent, one has

$$m_t = (1 - \alpha\lambda_t)\, \mu_{t|t-1} + \lambda_t(\tau_t - \bar{\tau}), \qquad \lambda_t \equiv \frac{\alpha\, v_{t|t-1}}{\alpha^2 v_{t|t-1} + \sigma_\eta^2},$$

so the signal-space threshold is affine:

$$\left| \tau_t - \bar{\tau} + \frac{1 - \alpha\lambda_t}{\lambda_t}\, \mu_{t|t-1} \right| > \frac{s^*}{\lambda_t}.$$

Under the normalization $\mu_{t|t-1} = 0$ (e.g., after any last adjustment the firm centers its prior), this reduces to $|\tau_t - \bar{\tau}| > \tau_t^*$ with $\tau_t^* \equiv s^*/\lambda_t$.

Proof. See Appendix A.

The Proposition says that the firm's optimal pricing rule takes a simple form: do nothing unless tips deviate enough from their average level to justify paying the menu cost. Specifically, the firm uses observed tips to form its best estimate of how far its current price is from the ideal price. If this estimated mispricing is small (within the "inaction band"), the firm leaves its price unchanged, because the cost of reprinting the menu outweighs the expected benefit of a better-aligned price. But if tips deviate sufficiently, either unusually high (suggesting underpricing) or unusually low (suggesting overpricing), the firm adjusts its price by exactly its best estimate of the gap.

The key insight is that this threshold can be expressed in terms of observable tips rather than unobservable mispricing. The width of the inaction band in tip space depends on two forces: the menu cost (which makes the band wider, since adjustment must clear a higher bar) and the informativeness of the tip signal (which makes the band narrower, since the firm can estimate mispricing more precisely). This is why the model predicts that firms with more informative tip signals, for example those with high transaction volumes that reduce sampling noise – will adjust prices more frequently.

2.1 Comparative Statics

The inaction band in s-space equals $s^* = \sqrt{2\kappa/\Lambda}$, so it widens with menu costs κ and narrows with demand curvature Λ. The tip-space threshold is $\tau_t^* = s^*/\lambda_t$, which shrinks when the signal is more precise (larger λ_t). From $\lambda_t = \alpha v_{t|t-1}/\left(\alpha^2 v_{t|t-1} + \sigma_\eta^2\right)$ we have $\partial\lambda_t/\partial\sigma_\eta^2 < 0$ and $\partial\lambda_t/\partial v_{t|t-1} > 0$: λ_t rises (i) when measurement noise falls (e.g., more transactions reduce sampling noise), and (ii) when prior uncertainty $v_{t|t-1}$ is higher, such as after longer periods without adjustment or when the efficient price is intrinsically more volatile (higher σ_ε^2). Thus the firm places more weight on the current tip signal when its prior is less precise.

The comparative statics can be summarized as follows. The firm is more likely to change its price when: (a) menu costs are low (cheaper to adjust), (b) demand is more price-sensitive (mispricing is more costly), (c) the tip signal is more precise (the firm can estimate mispricing more accurately), and (d) the firm has gone longer without adjusting (accumulated uncertainty makes even moderate tip deviations informative). Conversely, the firm tolerates larger tip deviations without acting when menu costs are high or when tips are noisy, for instance in settings where default tip options or social pressure obscure the surplus signal. In that sense, weaker or noisier tips make the pricing signal less decision-relevant, not irrelevant.

2.2 No Norms and Endogenous Mean Tipping with Search Frictions

As a robustness check on the role of the reference level $\bar{\tau}$ in (2), this subsection sketches an extension in which $\bar{\tau}$ is removed and the mean tipping rate is pinned down endogenously by customer mobility frictions. The goal is simply to show that the informational role of tipping – and the associated menu-cost pricing rule – does not hinge on taking an exogenous tipping norm as primitive.

In the baseline model, $\bar{\tau}$ is mainly a convenient benchmark against which observed tips can be compared. It is not the economic essence of the mechanism. In the extension, the firm no longer compares tips with an externally fixed norm. Instead, it learns from two observable facts: whether a transaction occurs at all and, conditional on a transaction, how high the realized tip is. No transaction is evidence that the posted price may be too high, while a higher realized tip after a transaction is evidence that the posted price may be too low. For that reason, the same menu-cost logic survives even when the benchmark is generated endogenously rather than imposed as a norm.

In each period t, a customer can either transact with the firm at the posted price p_t or search for an alternative option at a per-transaction search (switching) cost $\gamma > 0$. For simplicity, I capture this extensive-margin choice with a threshold rule:

$$\iota_t = \mathbf{1}\{s_t \geq -\gamma\}, \tag{5}$$

where $\iota_t = 1$ indicates that a transaction occurs (the customer stays) and $\iota_t = 0$ indicates that the customer searches away and no transaction occurs at the firm. The firm observes whether a transaction occurred (through sales), and conditional on $\iota_t = 1$ it also observes the realized tip rate τ_t.

Conditional on staying ($\iota_t = 1$), the customer leaves a nonnegative tip rate governed by a censored linear rule (with the same "informativeness" parameter α as before):

$$\tau_t = \max\{0,\ \alpha s_t + \eta_t\}, \qquad \eta_t \sim N\left(0, \sigma_\eta^2\right),\ \alpha > 0. \tag{6}$$

Equation (6) removes the intercept $\bar{\tau}$ entirely. Economically, tips remain informative about perceived surplus: higher s_t shifts the distribution of τ_t upward, but realized tips are bounded below by zero. The signal observed by the firm is now (ι_t, τ_t) rather than $\tau_t - \bar{\tau}$.

Let the firm's prior for mispricing be Gaussian as in the baseline, $s_t \sim N(\mu_{t|t-1}, v_{t|t-1})$. Because $\iota_t = 1$ implies $s_t \geq -\gamma$ and τ_t is censored at zero, the posterior distribution is no longer Gaussian in general. Nonetheless, the firm can summarize the information in the posterior mean and variance

$$m_t \equiv \mathbb{E}[s_t \mid \iota_t, \tau_t], \qquad v_t \equiv \mathrm{Var}(s_t \mid \iota_t, \tau_t),$$

and the one-period menu-cost logic derived above continues to apply: conditional on any posterior, the optimal adjustment size is $\Delta_t^* = m_t$ and the firm adjusts iff $\frac{1}{2}\Lambda m_t^2 > \kappa$.

A useful property in this extension is that, conditional on a realized *positive* tip, the posterior mean mispricing is strictly increasing in the observed tip. This delivers a threshold representation in tip space analogous to the linear–Gaussian case.

Lemma 1 (Posterior under positive tips and monotonicity). *Fix t and suppose $\iota_t = 1$ and $\tau_t = \tau > 0$. Let $\varphi(\cdot)$ and $\Phi(\cdot)$ denote the standard normal pdf and cdf. Then the posterior distribution of s_t is a truncated normal:*

$$s_t \mid (\iota_t = 1, \tau_t = \tau) \sim N(\tilde{m}_t(\tau),\ \tilde{v}_t) \text{ truncated to } [-\gamma, \infty),$$

where the (untruncated) Gaussian posterior parameters are

$$\tilde{v}_t \equiv \left(\frac{1}{v_{t|t-1}} + \frac{\alpha^2}{\sigma_\eta^2}\right)^{-1} = \frac{v_{t|t-1}\sigma_\eta^2}{\alpha^2 v_{t|t-1} + \sigma_\eta^2}, \tag{7}$$

$$\tilde{m}_t(\tau) \equiv \tilde{v}_t\left(\frac{\mu_{t|t-1}}{v_{t|t-1}} + \frac{\alpha\tau}{\sigma_\eta^2}\right) = (1 - \alpha\lambda_t)\mu_{t|t-1} + \lambda_t\tau, \quad \lambda_t \equiv \frac{\alpha\, v_{t|t-1}}{\alpha^2 v_{t|t-1} + \sigma_\eta^2} > 0. \tag{8}$$

Moreover, the posterior mean $m_t(\tau) = \mathbb{E}[s_t \mid \iota_t = 1, \tau_t = \tau]$ admits the closed form

$$m_t(\tau) = \tilde{m}_t(\tau) + \sqrt{\tilde{v}_t}\,\frac{\varphi(z_t(\tau))}{1 - \Phi(z_t(\tau))}, \qquad z_t(\tau) \equiv \frac{-\gamma - \tilde{m}_t(\tau)}{\sqrt{\tilde{v}_t}}, \tag{9}$$

and is strictly increasing in τ on $(0, \infty)$.

Proof. For $\tau > 0$ the censoring in (6) is inactive, so observing $\tau_t = \tau$ is equivalent to observing the linear Gaussian measurement $\tau = \alpha s_t + \eta_t$ together with the selection restriction $s_t \geq -\gamma$ implied by $\iota_t = 1$. Thus the posterior density satisfies

$$f(s \mid \iota_t = 1, \tau_t = \tau) \propto \exp\left(-\frac{(s-\mu_{t|t-1})^2}{2v_{t|t-1}}\right) \exp\left(-\frac{(\tau-\alpha s)^2}{2\sigma_\eta^2}\right) \mathbf{1}\{s \geq -\gamma\}.$$

Completing the square in s yields an (untruncated) normal kernel $N(\tilde{m}_t(\tau), \tilde{v}_t)$ with parameters (7) and (8). The indicator $\mathbf{1}\{s \geq -\gamma\}$ implies truncation at $-\gamma$, proving the distributional statement. The standard truncated-normal mean formula gives (9).

To show strict monotonicity, write $a \equiv -\gamma$, $\tilde{m} \equiv \tilde{m}_t(\tau)$, $\tilde{v} \equiv \tilde{v}_t$, $z \equiv (a - \tilde{m})/\sqrt{\tilde{v}}$, and define the inverse Mills ratio $r(z) \equiv \varphi(z)/(1 - \Phi(z))$. Then (9) can be written as $m = \tilde{m} + \sqrt{\tilde{v}}\, r(z)$. Differentiating with respect to $\tilde{m}$ gives

$$\frac{dm}{d\tilde{m}} = 1 + \sqrt{\tilde{v}}\, r'(z) \frac{\mathrm{d}z}{d\tilde{m}} = 1 - r'(z).$$

For a standard normal $Z \sim N(0,1)$ truncated below at z, one has $\mathrm{Var}(Z \mid Z \geq z) = 1 + z\,r(z) - r(z)^2 > 0$, hence $r(z)^2 - zr(z) < 1$ for all z. Using $r'(z) = r(z)^2 - zr(z)$ therefore gives $r'(z) < 1$ and so $dm/d\tilde{m} > 0$.

Finally, $\tilde{m}_t(\tau)$ is affine in τ with slope $\lambda_t > 0$ by (8), so

$$\frac{dm_t(\tau)}{d\tau} = \frac{dm}{d\tilde{m}} \cdot \frac{d\tilde{m}_t(\tau)}{d\tau} = (1 - r'(z_t(\tau)))\, \lambda_t > 0,$$

proving that $m_t(\tau)$ is strictly increasing on $(0, \infty)$. □

Lemma 1 implies that, conditional on $\iota_t = 1$ and $\tau_t > 0$, the menu-cost adjustment condition $\frac{1}{2}\Lambda m_t(\tau_t)^2 > \kappa$ can be written as a threshold rule in τ_t (since $\tau \mapsto m_t(\tau)$ is strictly monotone). For instance, under the common normalization $\mu_{t|t-1} = 0$ after an adjustment, there exists a cutoff $\tau_t^+ > 0$ solving $m_t\left(\tau_t^+\right) = s^*$ such that a price increase occurs iff $\tau_t > \tau_t^+$, where $s^* \equiv \sqrt{2\kappa/\Lambda}$. In contrast, downward adjustments are naturally tied to the extensive-margin event $\iota_t = 0$ (lost transactions), which shifts posterior mass toward $s_t < -\gamma$.

Because the extension contains no exogenous $\bar{\tau}$, define the endogenous mean tip rate among realized transactions as

$$\bar{\tau}^{\text{ endog}} \equiv \mathbb{E}[\tau_t \mid \iota_t = 1] \quad \text{under the stationary distribution induced by the pricing policy.}$$

Under (6), the conditional mean tip given mispricing s is available in closed form:

$$\mathbb{E}[\tau_t \mid s_t = s] = \sigma_\eta\, \varphi\left(\frac{\alpha s}{\sigma_\eta}\right) + \alpha s\, \Phi\left(\frac{\alpha s}{\sigma_\eta}\right), \tag{10}$$

so $\bar{\tau}^{\text{ endog}}$ is pinned down by primitives through the stationary distribution of s_t conditional on transacting ($s_t \geq -\gamma$). In particular, $\bar{\tau}^{\text{ endog}} > 0$ whenever $\sigma_\eta > 0$ and transactions occur with positive probability, even though the signal in (6) has no intercept.

For transparency, the remainder of the paper continues to work with the baseline linear–Gaussian signal (2), but the extension shows that the informational mechanism does not require an exogenous tipping norm. What changes in the extension is the source of the benchmark, not the pricing logic itself.

2.3 A Competitive Setting

Is it reasonable to assume that a firm always seeks to close the gap between its current price p and the notional target p^*?

We can justify this assumption by extending the single-firm framework to a competitive setting with two (or many) firms. For firm i, let $p_i^*(p_{-i})$ denote the static best-response price to currently posted rivals' prices p_{-i}; the observed tip rate continues to deliver a (noisy) signal of the firm-specific mispricing $s_i \equiv p_i^* - p_i$.

Under the same menu cost and quadratic-loss approximation used in this section, the optimal pricing rule is unchanged: a two-sided inaction region with adjustment to the posterior mean upon crossing the threshold. Rival price changes simply shift p_i^* (via strategic complements in standard differentiated-demand environments), so the interpretation of p^* as the efficient or best-response price is preserved and the qualitative implications (hazard rates, threshold comparative statics, and adjustment sizes) remain the same; the only new feature is occasional synchronization when large rival moves push p_i^* outside the inaction band. The N-firm case follows by replacing a single rival's price with p_{-i}. Intuitively, stronger competition raises own-price elasticity, which increases $\Lambda \equiv -\pi''(p^*)$ in the local quadratic loss and therefore shrinks the inaction band $s^* = \sqrt{2\kappa/\Lambda}$. Hence rivalry not only preserves the best-response interpretation of p^* but also tightens firms' tolerance for mispricing, strengthening the incentive to align p with p^*.

2.4 Dynamic Extension

The static analysis can be extended to a dynamic setting in which the efficient price p_t^* evolves stochastically and the firm updates its belief about the mispricing $s_t = p_t^* - p_t$ using the tip signal $\tau_t - \bar{\tau} = \alpha s_t + \eta_t$. The Kalman gain λ_t implies that the effective threshold in tip space, $\tau_t^* = s^*/\lambda_t$, declines when the firm has gone longer without adjusting or when tips are measured with less noise (for instance, on high–volume days). Consequently, the model predicts that the probability of a price change rises with time since the last adjustment and with the precision of the tip signal, and that the magnitude of the adjustment Δp_t increases in the absolute deviation of tips from their norm.

In the dynamic version, the firm faces an evolving target price and accumulates information from tips over time. Between price adjustments, the firm's uncertainty about its mispricing grows (because the efficient price keeps shifting), making the firm increasingly attentive to the tip signal. This creates a natural "pressure" toward eventual adjustment: the longer the firm waits, the more weight it places on tips, and the more likely it is that observed tips will push the posterior mean past the adjustment threshold. Conversely, immediately after an adjustment the firm is relatively confident that its price is well-aligned, so it discounts tip fluctuations as noise. This mechanism generates the prediction that the hazard of price adjustment increases with duration since the last change.

Figure 1 illustrates a simple impulse response: a one–time increase in p_t^* immediately raises observed tips, while the posted price adjusts only after the signal accumulates enough to cross the menu cost threshold, leaving a smaller residual gap thereafter. Notice that the impulse response shown corresponds to a sufficiently large shock to p_t^* such that the posterior mean m_t crosses the inaction threshold $|m_t| > s^*$, inducing a discrete price adjustment; smaller shocks would remain within the band and elicit no response from price.

Figure 2 plots a simulated path of the efficient price p_t^*, the posted price p_t, and the observed tip rate τ_t under a sequence of random shocks. The efficient price follows a random walk, while the firm adjusts its posted price only when the inferred mispricing exceeds the menu cost threshold. As a result, prices change infrequently

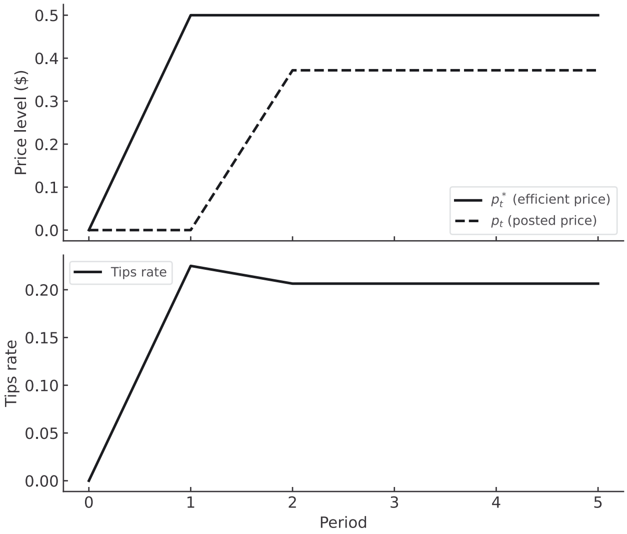

Figure 1: Impulse response to a one-time efficient-price shock. Solid: p_t^*; dashed: p_t; dotted vertical lines mark adjustment periods.

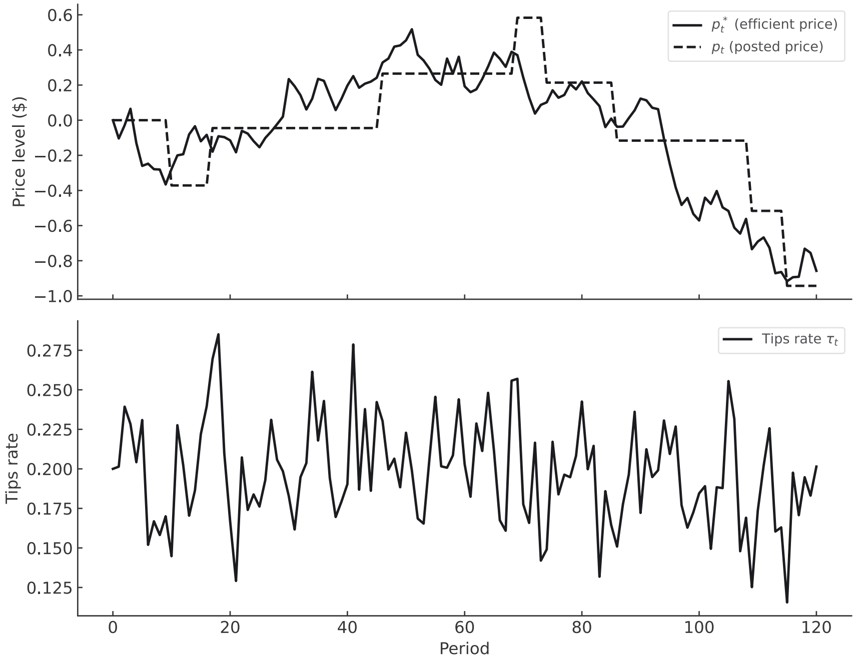

Figure 2: Simulated dynamics of the efficient price p_t^* (solid), posted price p_t (dashed), and tip rate τ_t (bottom panel). The mean $\bar{\tau}$ is set at 20 %.

and by discrete amounts, whereas tips move continuously with $p_t^* - p_t$ and remain persistently above or below their baseline between adjustments. The figure highlights how the model generates lumpy pricing, prolonged periods of inaction, and gradual information accumulation through tips – features consistent with real-world tipping behavior.

Details for the numerical illustration are provided in Appendix B.

3 Testable Implications and Comparative Statics

The model delivers sharp, falsifiable implications for the joint behavior of tips and posted prices. These implications are organized into time-series predictions (within a single firm over time) and cross-sectional predictions (across different types of firms).

Over time it implies: (i) the hazard of a posted-price change rises sharply once $|\tau_t - \bar{\tau}|$ crosses a threshold; (ii) the sign of the subsequent change matches the sign of the tip deviation; (iii) conditional on adjustment, the size of the change increases with $|\tau_t - \bar{\tau}|$.

Prediction (i) says that if a restaurant observes tips that are persistently and substantially above or below the norm, it should be more likely to change its menu prices soon. This distinguishes the model from pure norm-based explanations, which do not predict a systematic link between tip deviations and the timing of price changes. Prediction (ii) says that the direction of the price change should match the tip signal: above-normal tips (suggesting underpricing) should be followed by price increases, and below-normal tips by decreases. Prediction (iii) says that larger tip deviations should be followed by larger price adjustments. These predictions could in principle be tested using panel data on restaurant prices and tips, such as the data in Lynn and Ukhov (2013), who document that aggregate tip levels predict future restaurant sales, a pattern consistent with tips carrying the kind of pricing information the model formalizes.

The model also has cross-sectional implications across types of firms. Let σ_ε^2 denote volatility of the efficient price. Tipping is useful when σ_ε^2 is high (frequent mispricing), λ_t is high (tips are informative), and organizational menu costs are moderate. This is consistent with the casual observation that tipping is more common where service heterogeneity and demand shocks are large and less common where products and processes are standardized and centralized (quick-service chains).

The model predicts that tipping should be most prevalent in industries where three conditions coincide: (a) the "right" price shifts frequently (high σ_ε^2), so there is a recurring need for pricing information; (b) tips are relatively informative about customer surplus (high α, low σ_η^2), so the signal is worth attending to; and (c) the firm lacks cheaper alternatives for learning about mispricing, such as centralized point-of-sale analytics or frequent A/B testing. This explains why tipping is common in full-service restaurants (volatile demand, heterogeneous experiences, limited corporate analytics) but rare in fast-food chains (standardized products, centralized pricing algorithms, rich sales data). The model also predicts that tipping should decline, or become less decision-relevant to firms, as alternative feedback channels (e.g., online reviews, real-time demand data) become available, since these substitute for the informational role of tips. This is consistent with Lynn (2017)'s observation that the economic case for tipping varies substantially across restaurant formats.

4 Conclusions

Viewing tips as a pricing signal in a menu-cost environment produces a simple pricing rule and various empirical implications. The model does not rely on hard-to-measure social norms. The mechanism of the model suggests that, when a low-cost feedback channel exists, firms optimally economize on posted-price flexibility.

The framework may shed light on the growing practice of prompting customers for tips before service delivery. Although often criticized as a breakdown of traditional etiquette, this pattern is consistent with the model's informational logic: early tips still provide firms or platforms with information about perceived surplus and price adequacy, even before outcomes are realized. Seen through this lens, the expansion of pre-service tipping illustrates how the informational function of tips can persist even when their motivational role diminishes.

This paper presents only one possible explanation for the persistence of tipping behavior. The mechanism proposed here is that tips as an informational signal in the presence of menu costs, and it should be viewed as complementary to, rather than a replacement for, explanations based on social norms, reciprocity, or incentive considerations (the baseline specification takes the mean $\bar{\tau}$ as an exogenous reduced-form parameter, though Section 2.2 sketches one route to endogenizing the mean tip rate). Moreover, the model abstracts from the institutional details of how tips are distributed between owners and workers. In reality, the information content of tips may depend crucially on whether and how managers observe or share in the tipping revenue. A fuller treatment of these issues is left for future research.

Generative AI statement: During the preparation of this work, the author used ChatGPT 5.2 for copy-editing and producing graphs.
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Appendix A: Proof of the Proposition

Consider the firm's one-period problem conditional on the realized signal τ_t:

$$\min_{\Delta\in\mathbb{R},\ a\in\{0,1\}} \quad \mathbb{E}\left[\frac{1}{2}\Lambda\ (s_t - a\Delta)^2 |\ \tau_t\right] + a\ \kappa,$$

where $a = 1$ indicates that the firm adjusts and pays the menu cost (with choice of adjustment size Δ), and $a = 0$ indicates no adjustment (hence Δ is irrelevant and can be set to 0).

Step 1 (Optimal size conditional on adjusting). Fix $a = 1$ and minimize over Δ. Using the law of total expectation and writing $m_t = \mathbb{E}[s_t \mid \tau_t]$ and $v_t = \mathrm{Var}(s_t \mid \tau_t)$, we have

$$\begin{aligned}\mathbb{E}\left[(s_t - \Delta)^2 \mid \tau_t\right] &= \mathbb{E}\left[s_t^2 \mid \tau_t\right] - 2\Delta\, \mathbb{E}\left[s_t \mid \tau_t\right] + \Delta^2 = \left(m_t^2 + v_t\right) - 2\Delta m_t + \Delta^2 \\ &= (m_t - \Delta)^2 + v_t.\end{aligned}$$

Therefore the conditional objective given $a = 1$ is $\frac{1}{2}\Lambda\left((m_t - \Delta)^2 + v_t\right) + \kappa$, which is a strictly convex quadratic in Δ with unique minimizer at the first-order condition $-2(m_t - \Delta) = 0$, i.e. $\Delta_t^* = m_t$. This also implies the optimal direction: $\mathrm{sign}\left(\Delta_t^*\right) = \mathrm{sign}(m_t)$.

Step 2 (Value comparison: adjust vs. not). With $a = 0$ (no adjustment), setting $\Delta = 0$ yields expected loss $L_t^{\mathrm{no}} = \frac{1}{2}\Lambda\, \mathbb{E}\left[s_t^2 \mid \tau_t\right] = \frac{1}{2}\Lambda\, \left(m_t^2 + v_t\right)$.

With $a = 1$ and the optimal size $\Delta_t^* = m_t$, the expected loss is $L_t^{\mathrm{adj}} = \frac{1}{2}\Lambda\, \mathbb{E}\left[(s_t - m_t)^2 \mid \tau_t\right] + \kappa = \frac{1}{2}\Lambda\, v_t + \kappa$. Hence the *expected gain* from adjusting (gross of the menu cost) is $G_t \equiv L_t^{\mathrm{no}} - \frac{1}{2}\Lambda v_t = \frac{1}{2}\Lambda m_t^2$.

The firm prefers to adjust iff $L_t^{\mathrm{adj}} < L_t^{\mathrm{no}}$, i.e.,$\frac{1}{2}\Lambda m_t^2 > \kappa$. This is equivalent to $|m_t| > s^*$ with $s^* \equiv \sqrt{2\kappa/\Lambda}$. If $\frac{1}{2}\Lambda m_t^2 = \kappa$, the firm is indifferent between adjusting and not.

Step 3 (Signal-space representation). If the mapping $\tau_t \mapsto m_t$ is strictly monotone, the condition $|m_t| > s^*$ can be written as a two-sided threshold in τ_t via the inverse mapping. In the linear–Gaussian case with measurement $\tau_t - \bar{\tau} = \alpha s_t + \eta_t$, standard Bayesian updating yields

$$m_t = (1 - \alpha\lambda_t)\ \mu_{t|t-1} + \lambda_t\ (\tau_t - \bar{\tau}), \qquad \lambda_t = \frac{\alpha v_{t|t-1}}{\alpha^2 v_{t|t-1} + \sigma_\eta^2},$$

so the region of adjustment is

$$\left|\ (1 - \alpha\lambda_t)\mu_{t|t-1} + \lambda_t(\tau_t - \bar{\tau})\ \right| > s^*,$$

equivalently

$$\left|\ \tau_t - \bar{\tau} + \frac{1 - \alpha\lambda_t}{\lambda_t}\ \mu_{t|t-1}\ \right| > \frac{s^*}{\lambda_t}.$$

Under the common normalization $\mu_{t|t-1} = 0$ (e.g., after the last adjustment the firm centered its prior), this becomes $|\tau_t - \bar{\tau}| > \tau_t^*$ with $\tau_t^* = s^*/\lambda_t$.

Combining Steps 1–3 proves the proposition.

Appendix B: Dynamic Version

Let the efficient price follow a random walk $p_t^* = p_{t-1}^* + \varepsilon_t$ with $\varepsilon_t \sim \left(0, \sigma_\varepsilon^2\right)$ and define mispricing $s_t \equiv p_t^* - p_t$. Customers leave a percentage tip τ_t that provides a noisy signal $\tau_t - \bar{\tau} = \alpha s_t + \eta_t$ with $\eta_t \sim \left(0, \sigma_\eta^2\right)$ and $\alpha > 0$. Given a Gaussian prior $s_t \sim N(\mu_{t|t-1}, v_{t|t-1})$, Bayesian updating yields the posterior mean $m_t = (1 - \alpha\lambda_t)\mu_{t|t-1} + \lambda_t(\tau_t - \bar{\tau})$ and variance $v_t = (1 - \alpha\lambda_t)v_{t|t-1}$ where $\lambda_t \equiv \alpha v_{t|t-1} / \left(\alpha^2 v_{t|t-1} + \sigma_\eta^2\right)$ is the Kalman gain. Between periods $v_{t+1|t} = v_t + \sigma_\varepsilon^2$ and, after any adjustment, $\mu_{t+1|t} = 0$.

Proposition (optimal one-period rule). With a per-change menu cost $\kappa > 0$ and quadratic loss $\ell(s) = \frac{1}{2}\Lambda s^2$ ($\Lambda > 0$), the firm: (i) sets the *optimal adjustment size* $\Delta_t^* = m_t$ whenever it adjusts; (ii) *adjusts iff* $\frac{1}{2}\Lambda m_t^2 > \kappa$, i.e., $|m_t| > s^*$ with $s^* \equiv \sqrt{2\kappa/\Lambda}$. Under the linear–Gaussian signal, this is equivalent to a two-sided threshold in tip space $|\tau_t - \bar{\tau} + ((1 - \alpha\lambda_t)/\lambda_t)\mu_{t|t-1}| > s^*/\lambda_t$; under the common normalization $\mu_{t|t-1} = 0$, the rule is $|\tau_t - \bar{\tau}| > \tau_t^*$ with $\tau_t^* \equiv s^*/\lambda_t$.

Proof. Given τ_t, choose $a \in \{0, 1\}$ and $\Delta \in \mathbb{R}$ to minimize $E\left[\frac{1}{2}\Lambda(s_t - a\Delta)^2 \mid \tau_t\right] + a\kappa$. For $a = 1$, $E\left[(s_t - \Delta)^2 \mid \tau_t\right] = (m_t - \Delta)^2 + v_t$, so the strictly convex objective is minimized at $\Delta_t^* = m_t$. The associated loss is $L_t^{adj} = \frac{1}{2}\Lambda v_t + \kappa$. For $a = 0$, the loss is $L_t^{no} = \frac{1}{2}\Lambda\left(m_t^2 + v_t\right)$. Hence the gain from adjusting is $G_t = L_t^{no} - L_t^{adj} = \frac{1}{2}\Lambda m_t^2 - \kappa$; the firm adjusts iff $G_t > 0$, i.e., $|m_t| > s^*$. Because m_t is affine in τ_t, invertibility of the linear–Gaussian map implies the threshold representation in tip space above. □

Comparative statics. The inaction band in s-space $s^* = \sqrt{2\kappa/\Lambda}$ widens with menu cost and narrows with curvature. In tip space $\tau_t^* = s^*/\lambda_t$, and λ_t rises with prior variance and falls with measurement noise, $\partial\lambda_t/\partial v_{t|t-1} > 0$ and $\partial\lambda_t/\partial\sigma_\eta^2 < 0$, so the hazard of changing price increases with time since last adjustment and with transaction volume (better signal precision).

Numerical illustration (Figures 1 and 2). For the illustration I set $\alpha = 0.05$, $\sigma_\eta = 0.03$, $\sigma_\varepsilon = 0.10$, $\Lambda = 1$, and $\kappa = 0.045$ (so $s^* \approx 0.30$). A one-time $+0.5$ shock to p_t^* raises tips immediately by $\alpha\left(p_t^* - p_t\right)$; the posted price adjusts when $|m_t| > s^*$, leaving a small residual gap thereafter. The figure is generated by simulating the filter and threshold rule with $\eta_t \equiv 0$ to display the expected path. To illustrate how the model generates gradual information accumulation and discrete price adjustments, next I simulate a sequence of random shocks to the efficient price. Specifically, the efficient price follows a random walk $p_t^* = p_{t-1}^* + \varepsilon_t$, $\varepsilon_t \sim N\left(0, \sigma_\varepsilon^2\right)$, and the observed tip rate is $\tau_t = \bar{\tau} + \alpha\left(p_t^* - p_t\right) + \eta_t$, $\eta_t \sim$

$N\left(0, \sigma_\eta^2\right)$. The firm updates its belief about the mispricing $s_t = p_t^* - p_t$ using the Kalman filter $m_t = (1 - \alpha\lambda_t)\mu_{t|t-1} + \lambda_t(\tau_t - \bar{\tau})$ with $\lambda_t = \alpha v_{t|t-1} / \left(\alpha^2 v_{t|t-1} + \sigma_\eta^2\right)$ and $v_{t+1|t} = v_t + \sigma_\varepsilon^2$. It adjusts its price only when $|m_t| > s^* = \sqrt{2\kappa/\Lambda}$ and, if so, sets $\Delta p_t = \rho m_t$ with $\rho \leq 1$. The simulation begins in a steady state with $p_0 = p_0^*$ and $\tau_0 = \bar{\tau}$, and then iterates these equations forward for T periods using Gaussian draws for (ε_t, η_t). The resulting series for p_t^*, p_t, and τ_t display prolonged periods of inaction punctuated by discrete price changes – illustrating the lumpy adjustment patterns implied by the model.

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